

Program 4

Integrate Git with Docker for Source-Controlled Application Builds

Step1:

Create the directory program4 and within that create Dockerfile and [index.js](#) and execute command called npm init -y which will create the file called package.json

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~$ cd DevOps/
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ ls
program1  program2  program3
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ mkdir program4
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps$ cd program4
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ ls
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ nano index.js
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ nano index.js
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ npm init -y
Command 'npm' not found, but can be installed with:
sudo apt install npm

-----
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ npm init -y
Wrote to /home/1rv24mc051_karthikb/DevOps/program4/package.json:

{
  "name": "program4",
  "version": "1.0.0",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "description": ""
}

1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ ls
index.js  package.json
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ npm install express

added 68 packages, and audited 69 packages in 17s

18 packages are looking for funding
  run `npm fund` for details
```

Step2:

And install express also

Npm install express

Step3:

Open the browser and log in to your GitHub account and create a new repository. In the GitHub home page of your account click on the profile and go to Developer Settings and create a Personal Access Token

Again back to terminal and initialize the git locally → **git init**.

Add all the files which are present in that folder to the git using the command → **git add .**

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git add .
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git commit -m "Initial commit for Program 4 - Docker + Git Integration"
Author identity unknown

*** Please tell me who you are.

Run

    git config --global user.email "you@example.com"
    git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got '1rv24mc051_karthikb@karthik-Lenovo-G50-80.(none)')
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git config --global user.name "karthikb1010"
git config --global user.email "karthikb1010@gmail.com"
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git commit -m "Initial commit for Program 4 - Docker + Git Integration"
[master (root-commit) 24d22ae] Initial commit for Program 4 - Docker + Git Integration
 636 files changed, 72891 insertions(+)
 create mode 100644 Dockerfile
 create mode 100644 index.js
 create mode 100644 node_modules/.package-lock.json
 create mode 100644 node_modules/accepts/HISTORY.md
```

```
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git init
Reinitialized existing Git repository in /home/1rv24mc051_karthikb/DevOps/program4/.git/
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git add .
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git commit -m "Initial commit for Program 4 - Docker + Git Integration"
On branch main
nothing to commit, working tree clean
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git branch -M main
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git remote add origin https://github.com/karthikb1010/program4.git
error: remote origin already exists.
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git push -u origin main
Username for 'https://github.com': karthik1010
Password for 'https://karthik1010@github.com':
remote: Invalid username or token. Password authentication is not supported for Git operations.
fatal: Authentication failed for 'https://github.com/karthikb1010/program4.git/'
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git remote remove origin
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git remote add origin https://github.com/karthikb1010/program4.git
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$ git push -u origin main
Username for 'https://github.com': karthikb1010
Password for 'https://karthikb1010@github.com':
Enumerating objects: 719, done.
Counting objects: 100% (719/719), done.
Delta compression using up to 4 threads
Compressing objects: 100% (666/666), done.
Writing objects: 100% (719/719), 724.27 KiB | 4.31 MiB/s, done.
Total 719 (delta 144), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (144/144), done.
To https://github.com/karthikb1010/program4.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
1rv24mc051_karthikb@karthik-Lenovo-G50-80:~/DevOps/program4$
```

Step4:

Now open github and search program4

The screenshot shows a GitHub repository page for 'program4' by user 'karthikb1010'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main content area shows a commit history table with 1 commit. The commit is titled 'Initial commit for Program 4 - Docker + Git Integration' and was made 16 minutes ago. The commit includes files: node_modules, Dockerfile, index.js, package-lock.json, and package.json. The README section is visible below the commit history. The right sidebar contains sections for 'About', 'Releases', and 'Packages', all of which are currently empty.

File	Commit Message	Time
node_modules	Initial commit for Program 4 - Docker + Git Integration	16 minutes ago
Dockerfile	Initial commit for Program 4 - Docker + Git Integration	16 minutes ago
index.js	Initial commit for Program 4 - Docker + Git Integration	16 minutes ago
package-lock.json	Initial commit for Program 4 - Docker + Git Integration	16 minutes ago
package.json	Initial commit for Program 4 - Docker + Git Integration	16 minutes ago